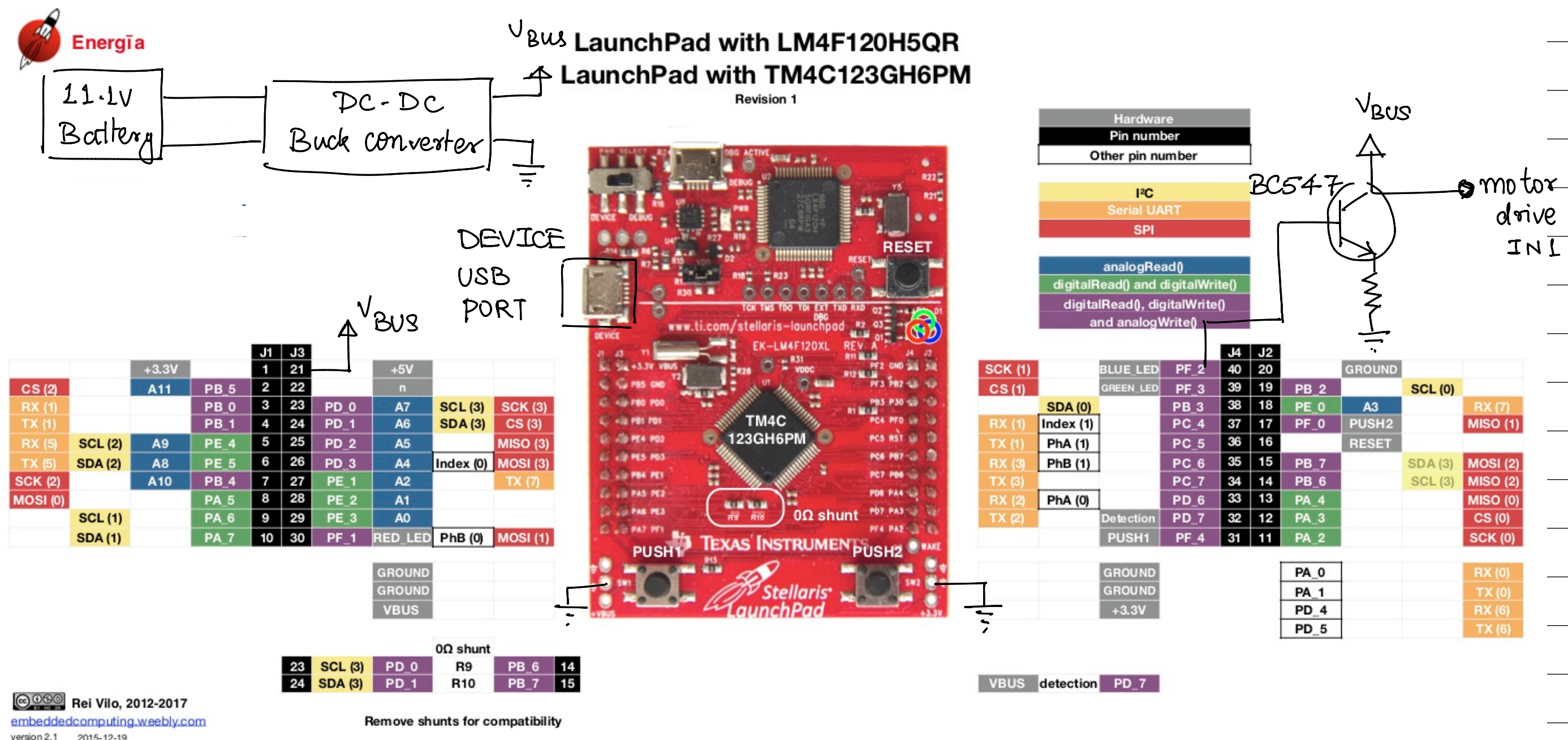


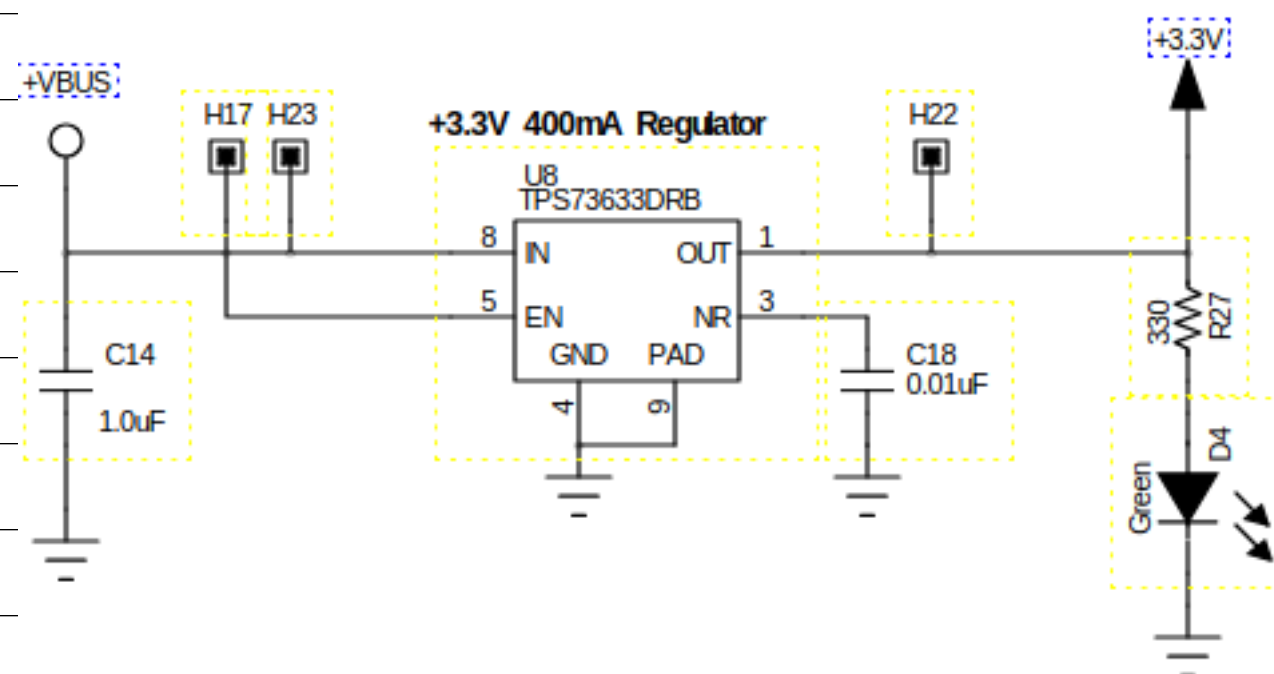
Tiva Board damage report :-

① Initially we were powering the board from the V_{bus} pin on the Tiva Board.



Buck converter was being tuned and ground was disconnected.

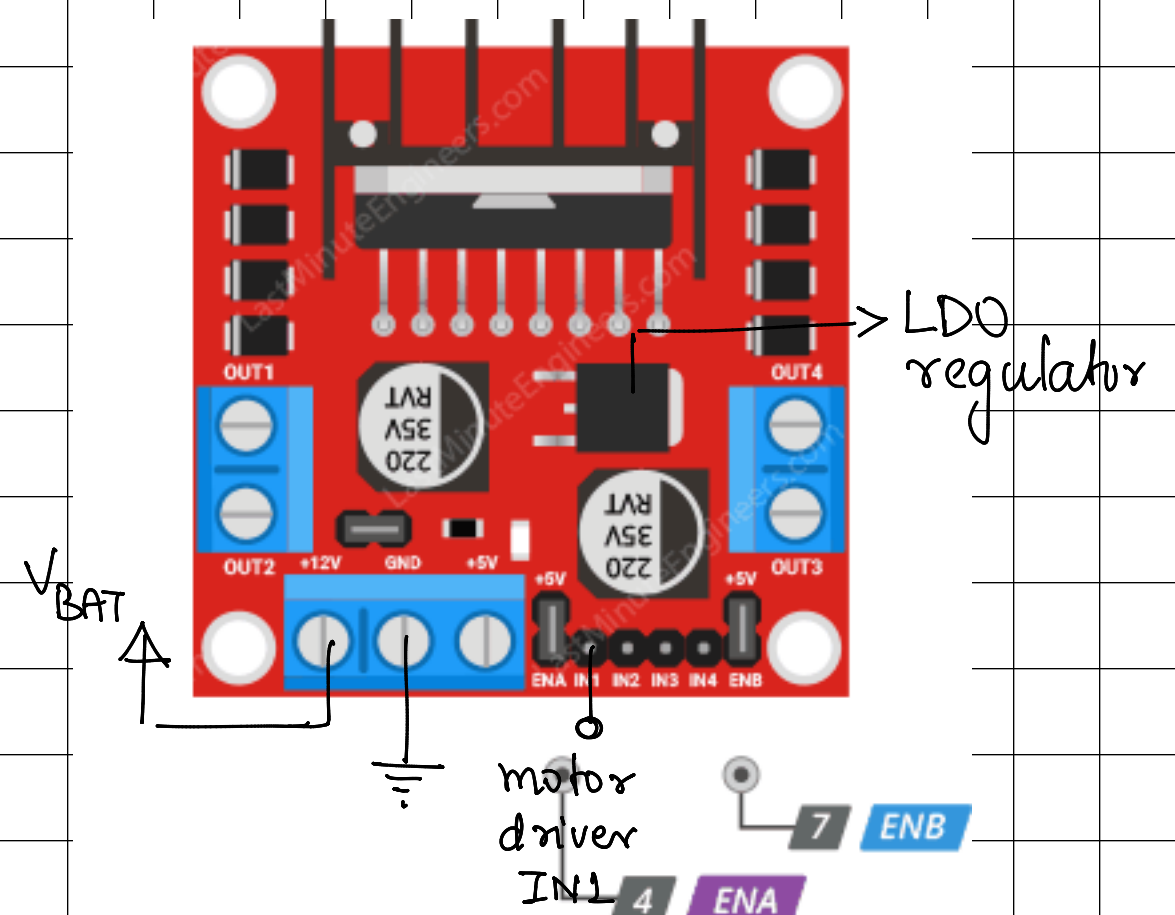
The reasoning for this electrical arrangement was that, in the board schematic a voltage regulator is mentioned which should convert +5V at V_{bus} to +3.3V at power pin of microcontroller.



<https://www.ti.com/lit/ug/spmu296/spmu296.pdf?ts=1764254346964>
 Tiva™ C Series TM4C123G LaunchPad Evaluation Board User's Guide

The reasoning for the BC547 circuit was to convert +3.3V pwm to +5V pwm, which would allow me to enable motor drive with LDO regulator on driver module, since it will have a common ground. This would have reduced the no. of wired connections from 3 per motor (EN, In_x, In_{x+1}) to 2 per motor (In_x, In_{x+1}).

However the board got damaged and the probable reason was some leakage current from PF2.



So the advice would be to properly tune the Buck converter before connecting any grounded peripheral.

Also instead of using Vbus pin use the "DEVICE" USB port, as it takes the electric power through some current limiting circuitry which should prevent the board from being fried.

To be on the safe side tune the Buck converter to just above +3.3V.

② Also protect the exposed pins on the TOP side from any metal parts to prevent them from being accidentally shorted.